
Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets its
2 refutation by overlapping authors; this paper joins them and locates the edge
3 of the set that refutation relies on. Conformal PID states long-run coverage for
4 any bounded scorecaster; a corollary keeps it for predictable perturbations of the
5 deployed threshold inside an admissible radius. The claim was acted on; placement
6 and derivation are conceded by name. At the null scorecaster that condition is *tight*.
7 A legal L_1 dead band on the *completed* threshold leaves that set at its radius, and
8 past it miscoverage is 1.000000 against $\alpha = 0.1$. Coverage goes, not only the rate.
9 The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere the
10 condition is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band
11 covers; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

1 Introduction

13 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,
14 by the same first and last authors, proves in the same notation that a bounded perturbation of the
15 deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the
16 joining is what this paper adds.

17 An online conformal method [Angelopoulos and Bates, 2021] emits a threshold before each obser-
18 vation and moves it. Forecast stability names, scores and prices that movement [Tunc et al., 2013,
19 Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026,
20 Van Belle et al., 2026]. Varying temporal structure at fixed level came earlier, in forecast verification,
21 matching predictive marginals on real producers [Gneiting et al., 2007, Pinson and Girard, 2012,
22 Worsnop et al., 2018]. Coverage and mean length under-describe a deployed interval [Min et al.,
23 2026, Vaze, 2026]. None of it is claimed here.

24 **The record carries a claim and its refutation; this paper joins them.** Conformal PID [An-
25 gelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be
26 predictable and lie in $[-b/2, b/2]$: “any function of the past” (Theorem 1). Such a readout is already
27 quantified over. The claim reads “[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical
28 coverage guarantee” [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, ver-
29 batim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2]
30 add any predictable d_t , $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ with $\mu_t \in [0, 1]$ predictable, to the deployed threshold;
31 since $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result “has exactly the CPID
32 error-integration form” and $|E_T| = o(T)$. The same generic-slot argument carries AcMCP’s long-run
33 coverage [Wang and Hyndman, 2024]. Both identifiers and “scorecaster” were searched across arXiv
34 full text, arXiv abstracts, OpenReview, and Semantic Scholar’s six citers of Li and Rodríguez [2025]:
35 none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice “arbitrary”
36 and lacking “principled guidance” while asserting valid coverage: model selection, not validity. **We**
37 **claim neither the placement nor the derivation.**

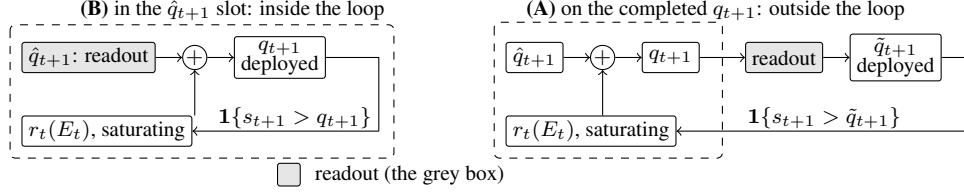


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In **(B)** the loop closes on the integrator’s own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster’s range, inside Corollary 2’s admissible set. In **(A)** it closes on $\tilde{q}_{t+1}$, which it never computed, and the deployed sequence leaves that set.

38 **Where that refutation stops, the condition is tight.** Corollary 2 is a *retention* result, silent outside
 39 its radius, and the question it leaves is whether that radius is sharp. Section 3 exhibits a legal
 40 perturbation leaving it: the L_1 dead-band family characterised here on the *completed* threshold,
 41 whose edge is the corollary’s radius at the null scorecaster. Past it coverage goes, not only the
 42 rate: the published sufficient condition is necessary there. Appendix C sets both claims beside
 43 four vocabularies for the same movement; Appendix D reads E_t as a capital process and builds a
 44 calibration test on it.

45 2 Setup

46 **The producer.** A forecaster fit before t induces a score s_t , and a threshold q_t is emitted before
 47 y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The
 48 target is long-run coverage, $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID
 49 [Angelopoulos et al., 2023] manipulates the threshold on the score scale, generalizing the running-
 50 error feedback that adaptive conformal inference [Gibbs and Candès, 2021] and its distribution-shift
 51 variant DtACI [Gibbs and Candès, 2024] already apply directly to the threshold:

$$q_{t+1} = \hat{q}_{t+1} + r_t \left(\sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

52 Angelopoulos et al.’s iteration (5). Its r_t saturates: condition (4) requires $|r_t(x)| \geq b$ with the sign of
 53 x once $|x| \geq c h(t)$, for h nonnegative, nondecreasing and sublinear. With $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$,
 54 Proposition 2 gives $|E_T| \leq c h(T) + 1$, read on realised scores.

55 **One scalar, two readouts, two places.** Damping a sequentially updated scalar has two standard
 56 readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1 - \lambda)\hat{q}_{t+1}^{\text{raw}} + \lambda\tilde{q}_t$
 57 [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied
 58 to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band,
 59 moving only when the target is more than τ away, then pulling back exactly τ [Constantinides,
 60 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; **(A)** is
 61 measured here. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form;*
 62 *it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits*
 63 *Proposition 2’s rate while covering, L_1 past the radius loses coverage, and at the legal $\hat{q} \equiv -b/2$ both*
 64 *lose it. What is tight is the radius, not a form.*

65 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*, $\mathcal{T}_H =$
 66 $\sum_{t=2}^H |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new. Over a predictive
 67 distribution $\sum |\Delta q|$ is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it
 68 *jumpiness*, a *convergence index*, (in)consistency [Zsótér et al., 2009, Ehret, 2010, Pappenberger
 69 et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length.
 70 Table 4’s caption names ten, so a fresh one is used here.

71 **Harness.** The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$,
 72 with $c = 1$ and $h(t) = \log(t + 2)$, meets (4) with equality. The scorecaster is null and legal, reducing
 73 (1) to the pure error integration Proposition 2 addresses. Neither choice is neutral: (4) is a lower
 74 bound met exactly here, and Theorem 1 permits scorecasters $b/2$ either side of null, predictable
 75 but not constant in t . The adversary is specified, not tie-broken. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$,
 76 $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the $\varepsilon \downarrow 0$ reading of “sets the score at the threshold”, firing the

77 strict indicator. Legality is Theorem 1’s $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight.
 78 *That specification names one member of a class:* any adapted $s_t \in [-b/2, b/2]$ is legal, the constant
 79 $s_t \equiv -b/2$ included. Every number in Table 2 (App. A) is measured against the one adversary above;
 80 Section 3’s two boundaries are quantified over the class, and they differ.

81 3 The edge of the admissible set, located and proved

82 **The retention result is published, and tested from outside.** Li et al. [2025, Corollary 2] prove
 83 that a predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration
 84 form and *retains* $|E_T| \leq ch(T) + 1$: the radius is theirs. A downstream partial adjustment of weight
 85 w leaves it; Table 2 (App. A) prices that departure, and five of its eleven arms exceed the bound at
 86 $T = 10^6$, and for $w = 0.99$ and 0.999 the excursion is *flat in T* while the bound grows like $\log T$.
 87 The forfeit is a property of the smoothing *strength*, not Placement A, and it buys something: travel
 88 falls from 28,311 to 202. A readout on the completed q_{t+1} touches neither r_t nor the integrator’s
 89 argument, so it cannot put (4) at risk. It delays a correction already computed, and the accumulator
 90 excurses *further*. Smoothing the output does not damp E_t ; it feeds it.

Table 1: Placement A, primary regime, $T = 10^6$: four arms of Table 2 (App. A), which lists the remaining seven and finer detail. Proposition 2’s bound is 14.8155; “ratio” is $\max_t |E_t|$ over it.

Arm	miscov. 10^6	$\max_t E_t 10^6$	ratio
none (control)	0.100007	7.80	0.53
partial adj. $w = 0.999$	0.100004	623.70	42.10
dead band $\tau = 0.9$	0.100012	13.20	0.89
dead band $\tau = 1.5$	1.000000	900,000	60,747

91 **The L_1 dead band leaves the radius, and the edge is a theorem.** A dead band pulls the deployed
 92 threshold back by exactly τ when it releases and holds it otherwise, so $|\hat{q}_{t+1} - q_{t+1}| \leq \tau$ at every
 93 round and “inside Corollary 2” reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value
 94 is pinned τ below the integrator’s ceiling, and its realised supremum is $b - \tau$ to the printed digit
 95 at every $\tau \geq 0.9$. Outside the radius two levels decide the answer and condition (4) names only
 96 one of them. Write $A_t^+ = \sup_x r_t(x)$ and $A_t^- = \inf_x r_t(x)$ for the integrator’s extremes, and
 97 $\Lambda_t^+ = \inf_{x \geq ch(t)} r_t(x)$, $\Lambda_t^- = \sup_{x \leq -ch(t)} r_t(x)$ for the levels it is *guaranteed* to reach once the
 98 accumulator clears the radius. Condition (4) says exactly $\Lambda_t^+ \geq b$ and $\Lambda_t^- \leq -b$; that $\Lambda_t^\pm = A_t^\pm$,
 99 that the integrator reaches its extremes where (4) puts them, is a further hypothesis and not one (4)
 100 supplies. Set

$$\tau^{\pm} = \sup_t (|A_t^\pm| \pm \hat{q}_{t+1}) - b/2, \quad \sigma^\pm = \inf_t (|\Lambda_t^\pm| \pm \hat{q}_{t+1}) - b/2. \quad (2)$$

101 **Failure, over the whole admissible class.** If $\tau > \tau^{++}$ then $\hat{q}_{t+1} + A_t^+ - \tau < b/2$ at every round,
 102 so $\tilde{q}_{t+1} \leq \max(\tilde{q}_t, q_{t+1} - \tau)$ makes $\{\tilde{q} < b/2\}$ absorbing: a release upward lands below $b/2$, and a
 103 hold or a release downward cannot lift the deployed value at all. Against $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ the
 104 indicator is then $1\{\tilde{q}_t < b/2\}$ and every round misses. If $\tau \geq \tau^{*-}$ the constant legal play $s_t \equiv -b/2$
 105 makes $\{\tilde{q} \geq -b/2\}$ absorbing by the mirror inequality, so no round can miss and the realised rate
 106 is 0 rather than α . Either way long-run coverage is lost, with $|E_T|$ linear in T . Nothing here asks $\hat{q}$
 107 to be constant, asks the extremes to be attained anywhere, or asks $A_t^\pm$ to be finite beyond what the
 108 hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If $\tau \leq \sigma^+$
 109 then $E_{t-1} \geq ch(t-1)$ forces $q_t \geq b/2 + \tau$ and so $\tilde{q}_t \geq b/2$, whence $\text{err}_t = 0$ against *every* legal
 110 adversary; if $\tau < \sigma^-$ the mirror forces $\text{err}_t = 1$ below $-ch(t-1)$. The accumulator cannot then
 111 leave $[-ch(t), ch(t)]$ by more than one round, and $|E_T| \leq ch(T) + 1$: Proposition 2’s own bound,
 112 retained verbatim rather than weakened to $o(T)$. Using condition (4) alone this already certifies
 113 $\tau \leq b/2 + \inf_t \hat{q}_t$ and $\tau < b/2 - \sup_t \hat{q}_t$, which for a constant scorecaster is Corollary 2’s radius at
 114 $\mu_t = 1$.

115 **The two directions meet, with no gap, on the sub-class the five settings inhabit.** If $\hat{q}_t \equiv \hat{q}$, if
 116 $\Lambda_t^\pm = A_t^\pm$, and if neither depends on t , then $\sigma^\pm = \tau^{\pm}$ and **coverage is retained against every legal**
 117 **adapted adversary if and only if $\tau \leq \tau^{++}$ and $\tau < \tau^{*-}$.** For a symmetric saturator that is one line:
 118 coverage if and only if $\tau \leq \sup_x r_t(x) - b/2 - |\hat{q}|$, with the endpoint included only when $\hat{q} < 0$.
 119 Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed**

120 **before is the failure half.** τ^{*+} is the rule Figures 2 and 3 (App. A) draw, and all 19 dead-band runs
 121 agree with it read against the adversary the harness plays. It is not a retention condition, and Table 3
 122 (App. A) separates the two with legal objects only. The supremum can sit where the dynamics never
 123 go: $|E_t| \leq (1 - \alpha)t$, so a saturator equal to the harness's inside $|x| \leq t^2$ and to $\pm 10b$ outside it
 124 satisfies (4), reads $\tau^{*+} = 19$, and fails at $\tau = 1.001$. A legal time-varying $\hat{q}$ splits $\sup_t$ from $\inf_t$:
 125 $\hat{q}_t = b/2$ on t a power of two and 0 elsewhere reads $\tau^{*+} = 2$ against $\sigma^+ = 1$. And τ^{*-} was missing
 126 entirely.

127 **The sharp form is one number, and its hypotheses are what buy it.** Specialise to a symmetric
 128 saturator that meets (4) with equality and attains its extremes there ($\Lambda_t^\pm = A_t^\pm = \pm b$; the harness's
 129 $r_t = b \text{ clip}(x/(ch(t)), \pm 1)$ is one, Section 2), and to a constant $\hat{q}$. Then

$$\min(\tau^{*+}, \tau^{*-}) = b/2 - |\hat{q}|, \quad (3)$$

130 with the endpoint admissible only when $\hat{q} < 0$: exactly Corollary 2's radius at $\mu_t = 1$, now as an
 131 if-and-only-if. **The necessity direction above assumes none of it:** $\tau > \tau^{*+}$, or $\tau \geq \tau^{*-}$, forfeits
 132 coverage against every admissible r_t and every legal predictable $\{\hat{q}_t\}$, extremes attained or not, and
 133 that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather
 134 than decorative: the unattained supremum and the varying- $\hat{q}$ split two paragraphs above are exactly
 135 what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they
 136 always do.

137 **The mirror boundary is already measured, on this paper's own grid.** The τ -cliff run's exact-
 138 threshold column under the strict indicator is the mirror adversary $s_t \equiv -b/2$ (one error process,
 139 $\text{err}_t = \mathbf{1}\{\hat{q}_t < -b/2\}$), and on the same eleven widths at the null scorecaster it locates $\tau^{*-} = 1$,
 140 closed on the failing side, exactly where (2) puts it; Appendix A prints the eleven readings. **So the**
 141 **failure criterion is $|E_T|$, not the miscoverage rate.** Both 1.000000 and 0.000000 are failures of the
 142 same order, $|E_T| = (1 - \alpha)T$ and αT ; and once $\hat{q}$ varies the rate stops being a tell at all.

143 **Both degenerate edges, and what they need.** An unbounded supremum is not on its own enough
 144 for $\tau^{*+} = \infty$ to mean what it looks like: the same unreachable-set construction with $r_t(x) = x$
 145 past $|x| > t^2$ has no finite supremum and fails at $\tau = 1.5$. What the tangent family has, and that
 146 construction does not, is a genuine reach: $\Lambda_t(X) = \tan(\arctan(b)X/(ch(t)))$, so the accumulator
 147 level at which the deployed threshold is forced to $b/2$ is $\kappa(\tau)ch(t)$ with $\kappa(\tau) = \arctan(b/2 +$
 148 $\tau - \hat{q})/\arctan(b)$, finite at every finite τ . Coverage is retained at every width, as measured; but
 149 $\kappa(\tau) > 1$ once $\tau > b/2 + \hat{q}$, so Proposition 2's own *constant* is forfeited past $\tau = b/2$ while coverage
 150 is not. At $\tau = 1.5$, $T = 10^6$ the excursion is 15.10 against Proposition 2's 14.8155 and against
 151 $\kappa(1.5)ch(T) + 1 = 15.8530$. At the other edge, $\sup_t \hat{q}_t = -b/2$ forces $\hat{q} \equiv -b/2$, since every $\hat{q}_t$
 152 is at least $-b/2$ already. There $q_{t+1} - \tau \leq -b/2 + b - \tau < b/2$ for every $\tau > 0$ and every t , and
 153 $\tilde{q}_1 = 0 < b/2$, so the failing set is absorbing from round 1 with no transient: the admissible window
 154 is the single point $\tau = 0$, and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$.

155 **What that establishes is that the published condition is tight.** At $\mu_t = 1$, $\hat{q} \equiv 0$ the boundary
 156 is exactly Corollary 2's radius, crossing it costs coverage rather than the rate ($\tau = 0.9$ covers at
 157 0.100012, $\tau = 1.5$ returns 1.000000 with $\max_t |E_t| = 900,000$, 60,747 times Proposition 2's bound
 158 at $T = 10^6$; Table 2, App. A), and (2) makes that an if-and-only-if against the whole legal adversary
 159 class rather than a pattern across five settings. **The claim is that necessity: a tightness result**
 160 **about someone else's sufficient condition**, smaller and more precise than a forfeit of the rate by
 161 post-processing. Elsewhere it stays sufficient only, and now provably so: under the tangent integrator
 162 Corollary 2 permits $\tau \leq b/2$ while no finite width forfeits coverage against any legal adversary,
 163 because the reach is genuine at every level. At $\hat{q} \equiv +b/2$ the looseness goes the other way and
 164 disappears: Corollary 2 permits radius 0, and (2) permits no width at all against the mirror adversary,
 165 the unsmoothed control included, though $\tau = 1.5$ covers there against the specified one. That
 166 asymmetry makes "tight" a located claim.

167 **The constructive reading, and it is the corollary's rather than ours.** Placement B keeps $\{\hat{q}_t\}$
 168 inside the admissible radius, so Theorem 1 carries it already. What Section 3 adds is that the band
 169 has two edges, both computable before deployment from the integrator's guaranteed reach and the
 170 scorecaster's range, that crossing either is not a graceful degradation, and that between $\sigma^\pm$ and $\tau^{*\pm}$,
 171 which separate only when $\hat{q}$ varies in t , neither direction is settled here.

4 Limitations

The retention direction is proved with no gap only where the two boundaries meet: a constant scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run here; a genuinely time-varying $\hat{q}$ leaves the band $\sigma^+ < \tau \leq \tau^{*+}$ open in neither direction (Section 3). And $0.63/(1-w)$ is specific to the saturator’s *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, the certified gap $(ch(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75b$. Under i.i.d. scores it returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. The i.i.d. check above used one seed** (20260820, $T = 10^6$); **the primary, adversarial harness uses none, and only its null-scorecaster edge is bracketed;** the others are one-sided. **What remains open is the band between the two boundaries.** For a genuinely time-varying $\hat{q}$, coverage in $\sigma^+ < \tau \leq \tau^{*+}$ is neither proved nor falsified; Table 3’s power-of-two rows sit there, and closing that band is future work, not a sweep this paper has already run.

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292 Code and data

293 The simulator, its frozen configuration, the results/ files behind every figure and table, the gen-
 294 erators that read them, and this project’s audit trail across sessions are at [https://\[REPOSITORY](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION])
 295 [LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION\]](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION]).

296 A Supporting measurements

297 **The primary regime.** Every number in Table 2 is measured under one deterministic adversary,
 298 which consumes no randomness and needs no seed. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$
 299 against the *deployed* threshold, and the indicator is strict. The specification is not a tie rule, and
 300 the distinction is measured rather than argued. Scoring an exact-threshold play as a *miscoverage*
 301 reproduces every quantity Table 2 reports: 44 of 44 arm-horizon cells identical, all eleven arms at
 302 all four horizons. Scoring the same play under the *strict* indicator does not, and it moves every
 303 row, because the adversary’s signature move then covers rather than misses. Partial adjustment at
 304 $w = 0.999$ excurses 70.80 in place of 623.70, and the failing dead band returns 0.000000 in place
 305 of 1.000000. The $\varepsilon \downarrow 0$ limit the harness specifies is the first reading, and that 623.70 turns on the
 306 adversary’s ε , not the counting convention. **No row is convention-free with respect to the second,**
 307 which is why the adversary is specified rather than tie-broken. At $\tau = \tau^{*+}$ itself the two readings part:
 308 the strict indicator returns 0.000000 where scoring the exact play as a miscoverage returns 1.000000.
 309 **Neither of those is coverage.** They are $|E_T| = \alpha T$ and $(1 - \alpha)T$, failures of the same order and
 310 opposite sign. Under the strict indicator the exact-threshold play scores $\text{err}_t = \mathbf{1}\{\tilde{q}_t < -b/2\}$,
 311 which is also the indicator of the constant legal play $s_t \equiv -b/2$, so that column is Section 3’s *mirror*

Table 2: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. A four-arm subset of this table – the control, the strongest partial-adjustment arm, and the two dead-band widths bracketing τ^{*+} – appears in the body as Table 1; this table adds the remaining seven arms, the $T = 2 \times 10^5$ horizon, saturation fraction, and threshold travel the body table omits. Proposition 2’s bound is 13.2061 at $T = 2 \times 10^5$ and 14.8155 at $T = 10^6$; “ratio” is $\max_t |E_t|$ over it, and “Sat.” the fraction of rounds on which (4) delivers its full $\pm b$. Every number is the *primary* regime’s, defined in this appendix, which also gives the $T = 10^4$ excursions and the fitted excursion law.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t E_t \cdot 10^6$	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	0.100004	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	1.000000	1.000000	900,000	60,747	1.0000	0.5

adversary rather than a second convention. It returns between 0.09992 and 0.09997 at $\tau = 0.5, 0.9$, 0.95, 0.99 and 0.999, and 0.000000 with $\max_t |E_t| = 10,000$ at $\tau = 1, 1.001, 1.01, 1.05, 1.1$ and 1.5. That locates $\tau^{*-} = 1$ on the same eleven widths Figure 2 uses, closed on the failing side. The convention decides which legal adversary realises the endpoint failure, not whether one occurs.

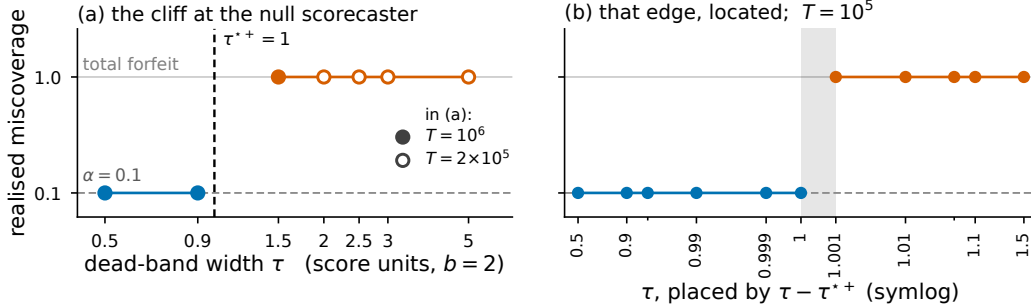


Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b , under the *specified* adversary, where the upward boundary τ^{*+} of (2) is 1. (a) The full sweep: filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2 \times 10^5$; blue covers, orange forfeits, and the dashed rule is τ^{*+} . (b) The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^{*+}$ on a symlog axis whose linear zone is $|\tau - \tau^{*+}| \leq 10^{-3}$, which magnifies the neighbourhood of τ^{*+} : visual spacing here is not proportional to true distance in τ . Unlabelled minor ticks mark $\tau = 0.95$ and 1.05 . The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four $(r_t, \hat{q})$ settings. The same eleven widths under the *mirror* adversary, which locates τ^{*-} , are read off in this appendix’s primary-regime paragraph.

Excursions at $T = 10^4$, and where the excursion law holds. Partial adjustment excurses 6.30, 63.00 and 623.70 at $w = 0.9, 0.99$ and 0.999 . The last two are flat in T while Proposition 2’s bound grows like $\log T$, which is why the ratio falls with the horizon: at $w = 0.999$ it is 61.08 at $T = 10^4$ and 42.10 at $T = 10^6$. On the control and the four partial-adjustment arms the excursion fits $\max_t |E_t| \approx \max\{0.5 h(T) + 0.9, 0.63/(1 - w)\}$ at every horizon, most loosely at $w = 0.9$, the one arm whose two branches cross inside the horizon range. This is a fit to those five arms, not a derived bound. It does not extend to the other six: the growing time constants, the running mean and the dead bands all excurse further than it predicts.

Realised miscoverage at the covering settings of Figure 3. At $\hat{q} \equiv +b/2$ the $\tau = 1.5$ band returns 0.100010 with $\max_t |E_t| = 11.20$, inside the bound. Under Angelopoulos et al.’s tangent integrator the same width returns 0.100001. Raising the saturator’s level to $4b$ leaves coverage intact and cuts

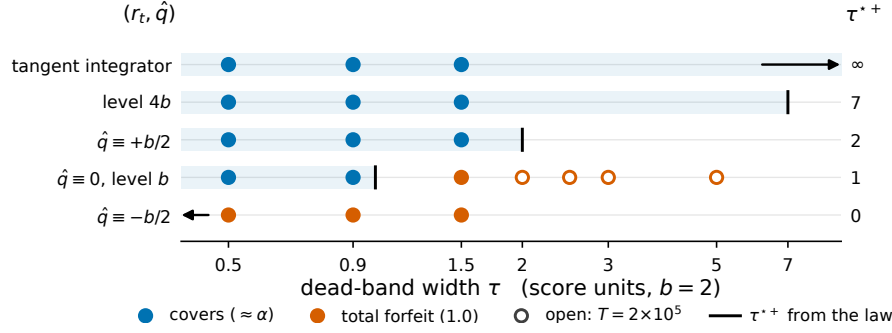


Figure 3: All 19 dead-band runs, at the five admissible $(r_t, \hat{q})$ settings: rows are settings ordered by τ^* , and the shared horizontal axis is τ . Blue markers cover at $\approx \alpha$ and orange markers return 1.000000; filled markers run to $T = 10^6$ and open markers to $T = 2 \times 10^5$, all under the *specified* adversary. The rule on each row is that setting's upward boundary $\tau^{*+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, and the tinted band is $\tau \leq \tau^{*+}$. Two rules fall off the axis and are drawn as arrows: $\tau^{*+} = 0$ at $\hat{q} \equiv -b/2$, and τ^{*+} unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary τ^{*-} of (2) is not drawn here and differs from τ^{*+} at the two non-null scorecasters.

the $w = 0.999$ excursion from 623.70 to 120.60. That is the first term of τ^* moving the edge, not the readout changing behaviour.

What survives of the measurement, and what its status now is. Outside the admissible window the transition is a step. Past τ^{*+} the threshold sits permanently below the reachable score range and $|E_t| = (1 - \alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$; the wider bands $\tau = 2.0, 2.5, 3.0$ and 5.0 fail too at $T = 2 \times 10^5$ (Figure 2). **Here** $\sup_x r_t(x) = b$ and $\hat{q} \equiv 0$, so $\tau^{*+} = \tau^{*-} = b/2$, one point of the admissible class, and both terms move it. The legal $\hat{q} \equiv +b/2$ carries the $\tau = 1.5$ band inside the bound against the specified adversary, and against the mirror one no width covers there, the unsmoothed control included. Condition (4) bounds $\sup_x r_t(x)$ only from below, so raising it costs the failure direction nothing; it moves the edge too (Figure 3) and shrinks the excursion, from 623.70 to 20.10 under the tangent integrator Angelopoulos et al. [2023] report using in all their experiments, both at $T = 10^6$. **The method's own integrator therefore sits inside the set at every width run here. The endpoint is where the two boundaries part.** The upper one is closed and the lower one is open, which the null scorecaster cannot show because τ^{*+} and τ^{*-} coincide there: four saturators with $\tau^{*+} < \tau^{*-}$ cover at $\tau = \tau^{*+}$ exactly under both adversaries, and at $\hat{q} \equiv +0.4$, where $\tau^{*-} = 0.6 < 1.4 = \tau^{*+}$, the width 0.599 covers under both while 0.6 fails under the mirror. At the baseline the two coincide at 1, so the primary regime covers there and the mirror adversary does not; Figure 2 and this appendix's primary-regime paragraph print both readings. **The failing arms have** Sat. = 1.0000. Condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2's other hypothesis: the threshold closing the loop is the integrator's output.

Where the rate stops being a tell. At $\tau = 1.9$ with the power-of-two scorecaster the rate is 0.110960 against $\alpha = 0.1$ while $\max_t |E_t|$ is 84,745.70 at $T = 10^6$, so a reader watching only the rate would call that coverage.

Table 3: Legal $(r_t, \hat{q})$ configurations on which the printed τ^* over-certifies, each under two legal adversaries. “Proved window” is where (2) certifies retention against every legal adversary; between it and the failure boundaries $\tau^{*\pm}$ neither direction is claimed, and the rows built on a supremum the dynamics cannot reach or never attain sit there. Here $\emptyset$ marks a configuration for which no width is certified, $\tau = 0$ included. $T = 10^5$ except the two power-of-two rows, at $T = 10^6$. The last row carries no dead band at all.

$(r_t, \hat{q})$ configuration	τ	printed τ^*	proved window	$s_t = \tilde{q}_t + \varepsilon$		$s_t \equiv -b/2$	
				miscov.	$\max_t E_t $	miscov.	$\max_t E_t $
$\hat{q} \equiv +0.4$, level b	0.599	1.4	$\tau < 0.6$	0.100010	7.20	0.099890	11.60
$\hat{q} \equiv +0.4$, level b	0.8	1.4	$\tau < 0.6$	0.100050	8.70	0.000000	10,000
$\sup_x r_t = 10b$, off the reach	1.001	19	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t$ unbounded, off the reach	1.5	∞	$\tau < 1$	1.000000	90,000	0.000000	10,000
$A^+ = 4b$, $A^- = -b$	1.5	7	$\tau < 1$	0.100020	4.30	0.000000	10,000
level $4b$, $\hat{q} \equiv 0$	6.999	7	$\tau < 7$	0.100070	12.20	0.100050	11.60
level $4b$, $\hat{q} \equiv 0$	7.0	7	$\tau < 7$	0.100090	12.20	0.000000	10,000
$\sup_x r_t = 2b$, unattained	2.99	3	$\tau < 1$	0.100320	211.10	0.098010	210.50
$\sup_x r_t = 2b$, unattained	3.0	3	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t = 2b$, attained	3.0	3	$\tau < 3$	0.100000	12.20	0.000000	10,000
$\hat{q}_t = b/2$ on powers of 2	1.0	2	$\emptyset$	0.100009	14.30	0.000000	100,000
$\hat{q}_t = b/2$ on powers of 2	1.9	2	$\emptyset$	0.110960	84,745.70	0.000000	100,000
$\hat{q} \equiv +b/2$, no band	0	2	$\emptyset$	0.100000	0.90	0.000000	10,000

B The four vocabularies

Table 4: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness, convergence index, (in)consistency*; and *switching cost*. The row gives what each contributes.

Online conformal	Forecast stability	Verification	Switching-cost online learning
Distribution-free long-run coverage for any bounded scorecaster as post-processing [Angelopoulos et al., 2023, Thm 1, Prop. 2]	Movement functional; readout as post-processing [Godaheva et al., 2025, Eq. 3]; priced decision [Genov et al., 2026, §4.2], [Van Belle et al., 2026]	Matched-level design, real producers (2012) [Pinson and Girard, 2012, Worsnop et al., 2018]	Worst-case theory for the penalised problem [Andrew et al., 2013, Thm 2]

C Where this sits

Four vocabularies name one quantity, and Table 4 names all four; this appendix places this paper’s claim and its concessions among them. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves; this paper proves that admissible set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID’s own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t \leq T} q_t \leq O(\alpha h(T))$ for a sublinear h , supported with a plot rather than derived. Verification’s matched-level design tests real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a Thm. 7 trade-off in which sublinear regret is bought by growing θ with T , which grows the ratio [Andrew et al., 2013, Thm. 7].

The accumulator, read as a bet. Appendix D reads E_t as a game-theoretic capital process and builds a calibration test on top of the boundary above; it is not repeated here.

D The accumulator as a bet, and a calibration test built on the boundary

What is assumed here, and by whom. Everything Section 3 proves is deterministic and pathwise. Proposition 2’s window $|E_T| \leq c h(T) + 1$, and the boundaries (2) that locate its edge, hold on every trajectory a legal adapted adversary can produce; no distributional model on the data appears in either statement, and neither is a statement about a null hypothesis. This section neither uses that nor weakens it. It runs the other way: it takes the error sequence a deployment actually produced as *data*, and tests a hypothesis about it. The hypothesis is imposed here, for the test alone, with $\mathcal{F}_{t-1}$ the history through round $t - 1$ (scores, deployed thresholds and errors) against which $\hat{q}_t$ is already required to be predictable:

$$H_0 : \quad \text{err}_t \mid \mathcal{F}_{t-1} \sim \text{Bernoulli}(\alpha) \quad \text{for every } t,$$

and it says that the shipped intervals were conditionally calibrated at the target rate. H_0 is the object under test. It is not a premise of Proposition 2, of (2), or of any other claim in this paper; nothing above requires it, implies it, or is disturbed if it fails. When it fails, what loses validity is the test below and nothing else. We are explicit about this because the rest of the paper assumes no distribution at all, and a reader who meets a Bernoulli null on this page must not read it backwards into the pages before it.

The accumulator is a capital process, and it is signed. A Skeptic who each round pays α for the payoff err_t has cumulative net gain exactly E_t , so this paper’s accumulator is a game-theoretic capital

process at unit stake [Shafer and Vovk, 2019, Ramdas et al., 2023]. It is not an e-value and not an e-process: it is signed, it goes negative on exactly the trajectories the mirror boundary τ^{*-} governs, and Proposition 2 bounds it pathwise by feedback rather than under a null by Ville’s inequality [Vovk and Wang, 2021]. Nothing below proposes E_t , or a transform of it, as a certificate for any claim this paper makes. What is formal is the standard transform, and it is invoked rather than invented.

The test martingale, and why it is one. Fix λ and set $M_0 = 1$,

$$M_t = \prod_{i \leq t} (1 + \lambda(\text{err}_i - \alpha)).$$

Each factor takes one of two values, $1 + \lambda(1 - \alpha)$ on a miss and $1 - \lambda\alpha$ on a cover, so $M_t \geq 0$ for every path if and only if $\lambda \in [-1/(1 - \alpha), 1/\alpha]$, which for $\alpha = 0.1$ is $[-10/9, 10]$. Under H_0 , $\mathbb{E}[\text{err}_t | \mathcal{F}_{t-1}] = \alpha$, so each factor is conditionally mean one and $\mathbb{E}[M_t | \mathcal{F}_{t-1}] = M_{t-1}$: M_t is a nonnegative martingale started at 1, i.e. a test martingale for H_0 . Ville’s inequality then gives $\mathbb{P}_{H_0}(\sup_t M_t \geq 1/\eta) \leq \eta$, so “reject H_0 the first time $M_t \geq 1/\eta$ ” is an anytime-valid level- η test of the deployed intervals’ calibration, at any stopping time and with no horizon fixed in advance. That is a test of the deployment, not of Section 3.

The boundary transfers exactly, because M_t sees only (t, E_t) . Since $\text{err}_i \in \{0, 1\}$, the product collapses. Writing N_t for the miss count and using $N_t = \alpha t + E_t$,

$$\begin{aligned} \log M_t &= K(\lambda)t + L(\lambda)E_t, \\ K(\lambda) &= \alpha \log(1 + \lambda(1 - \alpha)) + (1 - \alpha) \log(1 - \lambda\alpha), \\ L(\lambda) &= \log \frac{1 + \lambda(1 - \alpha)}{1 - \lambda\alpha}. \end{aligned}$$

with $K(\lambda) \leq 0$, $K(\lambda) = 0$ only at $\lambda = 0$, and $\text{sign } L(\lambda) = \text{sign } \lambda$. So M_t depends on the path only through the horizon and the accumulator – the same statistic Section 3’s dichotomy is stated in – and the transfer is an identity rather than the first-order expansion $\log M_t = \lambda E_t + O(\lambda^2 t)$. It transfers as a dichotomy, not as a measured rate. Inside the retained band $|E_t| \leq c h(t) + 1$ is logarithmic, so $\log M_t \leq K(\lambda)t + |L(\lambda)|(c h(t) + 1)$ at every round. Since $K(\lambda) < 0$ the right side falls to $-\infty$ linearly, so $M_T \rightarrow 0$ exponentially; and since it is continuous with a single interior maximum it has a finite supremum $S(\lambda)$ over all t , so $\sup_t M_t \leq e^{S(\lambda)}$ uniformly in the horizon. The test therefore never rejects at any level $\eta < e^{-S(\lambda)}$, however long the deployment runs. It is the uniform bound and not the limit that non-rejection needs, because Ville’s inequality prices the supremum. Past either boundary $|E_T|$ is linear in T , so some fixed λ of the sign of E_T makes $\log M_T$ grow linearly and the test rejects at any fixed level within $O(1)$ rounds. Both signs are needed, and that is the point of the two-sided form: the τ^{*+} failure drives E_T up and calls for $\lambda > 0$, the mirror failure at τ^{*-} drives it down and calls for $\lambda < 0$.

Worked on this paper’s own frozen arms, at the null scorecaster. Take the harness of Section 2 ($\alpha = 0.1$, $b = 2$, $c = 1$, $h(t) = \log(t + 2)$, $\hat{q} \equiv 0$), where $\tau^{*+} = \tau^{*-} = 1$, and the two dead bands that bracket that edge in Table 2 (App. A). *Past the edge*, $\tau = 1.5$ returns miscoverage 1.000000 and $\max_t |E_t| = 900,000$ at $T = 10^6$; the run records $n_{\text{err}} = T$ at all four horizons, so the per-round path is exactly $\text{err}_t = 1$ and $E_t = (1 - \alpha)t$, and no reconstruction is involved. A genuine pre-registered deployment fixes λ , sign included, before the data arrive – by running both signs as one two-sided test, or by a mixture over λ – and the single sign used per arm below is chosen after the fact, to illustrate the identity on a trajectory already known to fail or to hold; it is not itself a claim of a pre-registered test. At $\lambda = 1$ – a round number well inside $[-10/9, 10]$, not the aggressive endpoint – the identity above gives $M_t = 1.9^t$, so at level $\eta = 0.05$, threshold $1/\eta = 20$: $M_4 = 13.03$ and $M_5 = 24.76$. **The anytime-valid test rejects on the fifth deployed interval.** At $\eta = 0.001$ it rejects on the eleventh ($M_{10} = 613.11$, $M_{11} = 1164.90$). *Inside the edge*, $\tau = 0.9$ covers at 0.100012 with $\max_t |E_t| = 13.20$ against Proposition 2’s 14.8155 at the same $T = 10^6$, and the same $\lambda = 1$ gives $\log M_t \leq -0.030639t + 9.8632$ from the measured excursion alone: below $\log 20$ from round 225, below 0 from round 322, and $M_T \leq e^{-30,629}$ at $T = 10^6$. The measured excursion bounds one run; the theorem bounds every one. This width sits inside $\sigma^\pm$, so the retention half applies and $|E_t| \leq c h(t) + 1$ holds at every round, whence $S(1) = \sup_t (K(1)t + L(1)(\log(t + 2) + 1)) = 2.448$, attained at $t = 22$, and $M_t \leq 11.57$ at every round and every horizon. The test cannot reject at

435 $\eta = 0.05$ – not merely after round 225, but ever. Five rounds against never, on two arms of one
 436 frozen grid. *The mirror direction*, on the same $\tau = 1.5$ configuration read under the strict indicator
 437 against $s_t \equiv -b/2$: miscoverage 0.000000, $\text{err}_t = 0$ at every round, $E_t = -\alpha t$, and $\lambda = -1$ gives
 438 $M_t = 1.1^t$, which crosses 20 at $t = 32$ ($M_{31} = 19.19$). The asymmetry is not a defect of the test: a
 439 forced miss carries $\log 1.9$ nats per round against a forced cover’s $\log 1.1$, because $\alpha = 0.1$ makes
 440 misses the rarer event under H_0 .

441 **What this adds, and what it does not.** It adds a diagnostic and a reading, not a result. The
 442 reading is that E_t is the Skeptic’s capital and that Section 3 therefore locates, in a quantity a bettor
 443 already cares about, the exact width at which a dead band turns a calibrated deployment into one that
 444 an anytime-valid test convicts within a handful of rounds. The diagnostic is runnable online by a
 445 practitioner who has the errors and not the proof. Neither statement is a claim about E_t ’s status: E_t is
 446 not an e-process, M_t is a textbook transform imposed on top of the result rather than a part of it, and
 447 this paper’s contribution is the tightness of someone else’s admissible radius, proved pathwise. M_t is
 448 itself an e-value, and we say so: an e-value for H_0 alone, a hypothesis about one deployment’s error
 449 sequence, imposed on this page and nowhere else in the paper. No result of this paper is an e-value
 450 result or rests on one. Betting enters online conformal inference elsewhere as an optimiser rather
 451 than as evidence [Podkopaev et al., 2024].